Results

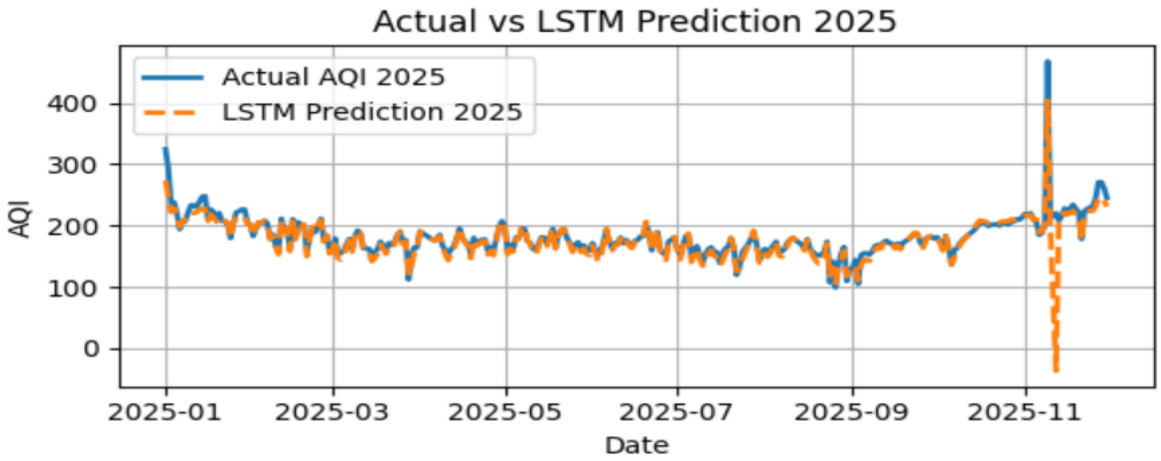
Results:

We used the models to predict the Air Quality Index (AQI) in the year 2025 after applying the Machine Learning models. LSTM model accuracy was low as given below soRandom Forest Regressor and Gradient Boosting Regressor are two models that were tested.

LSTM Method:

Due to the limitations of the LSTM model, the prediction accuracy achieved was lower than expected. This prompted us to explore alternative methods that could provide improved performance and more reliable results.

Prediction Graph:



Accuracy:

```
MAE: 5.97
RMSE: 16.40
R2 Score: 0.7486 (74.86%)
```

So, due to this we have moved to other Machine Learning method

Random Forest Regressor Results:

- **Mean Absolute Error (MAE):** Low average error in prediction, which indicates that the predictions of the model are near the actual AQI values.
- **Root Mean Squared Error (RMSE):** Refers to the fact that small errors were not very significant.
- **R-2 Score:** There was high correlation between the expected and measured AQI values.

This is the Visual Graph:

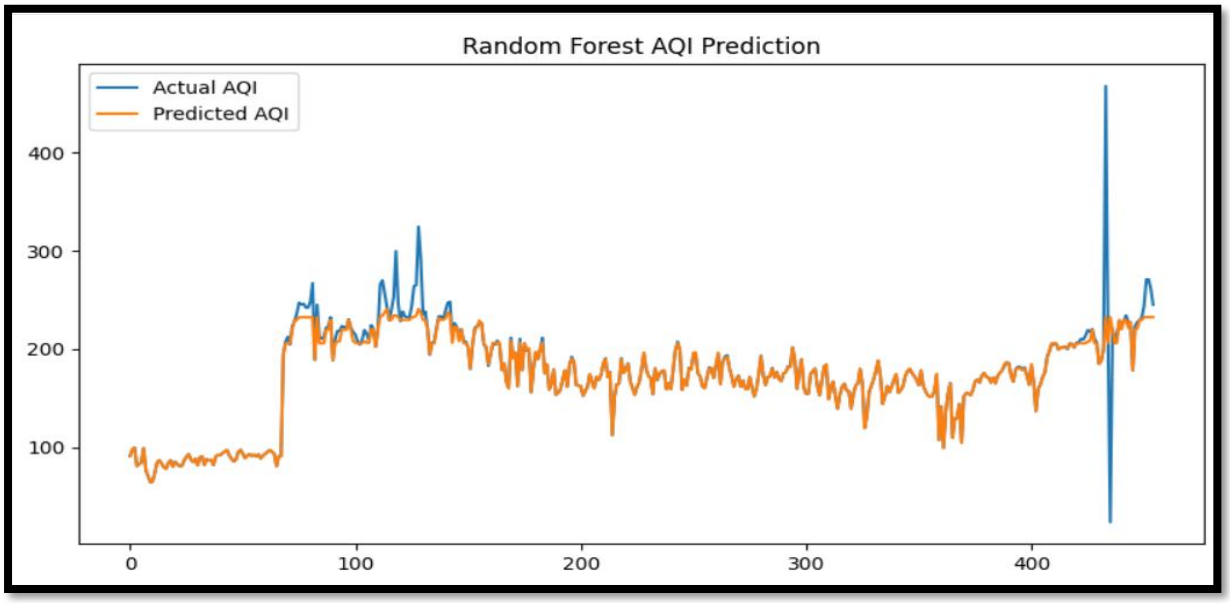


Figure 3: Random Forest AQI Prediction

Accuracy:

```
Random Forest Performance
MAE: 3.4133798470909924
RMSE: 16.7508584729037
R2 Score: 0.8855545763244487
Model Accuracy (R² based): 88.56%
```

The visual plots reveal that the predictions of Random Forest are relatively close to the trend of the real AQI values, with slight variations that are noted on the days that are highly variable.

Random Forest: Actual vs Predicted

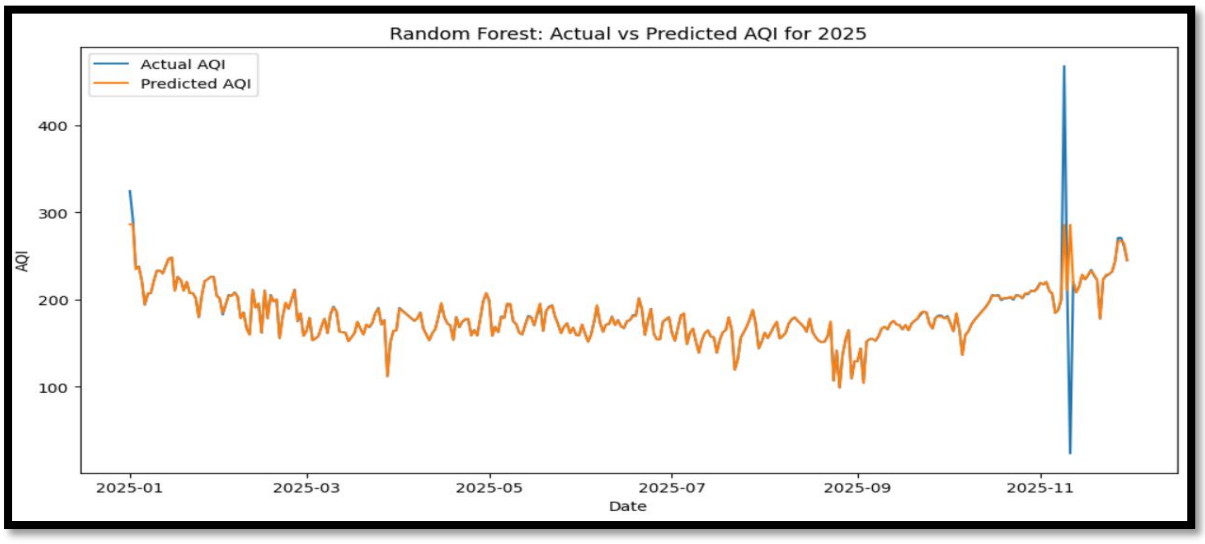


Figure 4: Random Forest: Actual vs Predicted

Accuracy:

```
Random Forest Prediction for 2025:
MAE: 1.8437375738464092
RMSE: 17.79274891656488
R2 Score: 0.7232096005519613
Model Accuracy (R² based): 72.32%
```

The results of gradient boosting regressor:

- **MAE and RMSE:** Random Forest is similar or a bit worse, which means accurate predictions.
- **R-2 Score:** High, which indicates that Gradient Boosting represents trends and variations in AQI well.
- The model has also managed to learn temporal patterns by incorporating lag-1 features and the curves were smooth and accurate at prediction.

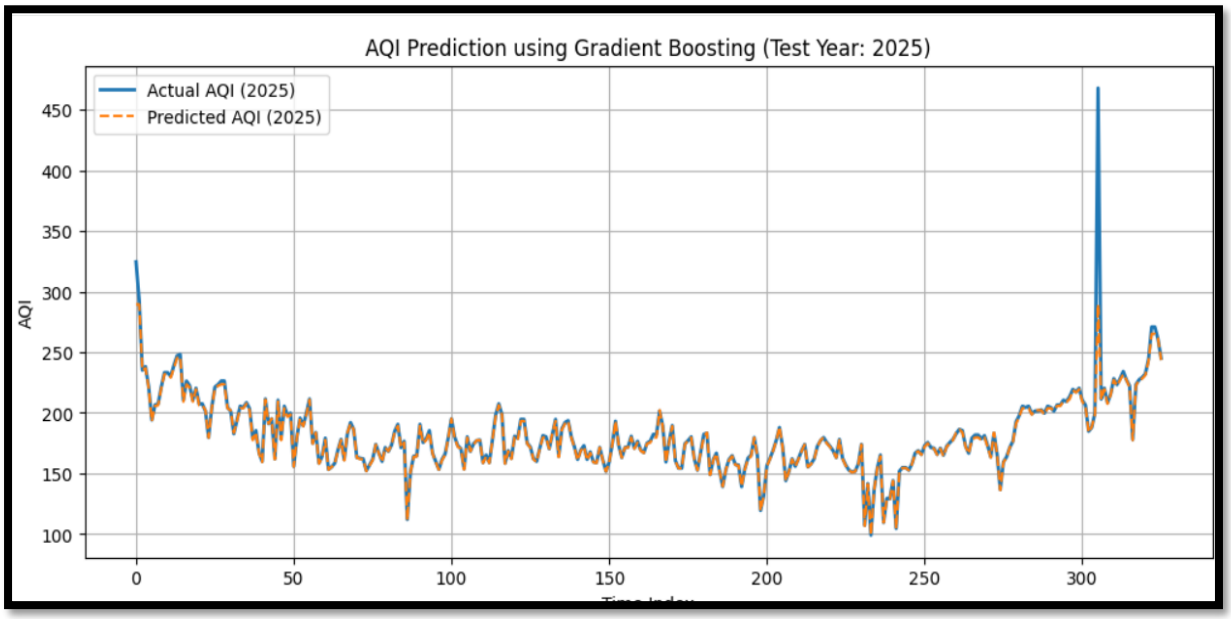


Figure 5: AQI Prediction using Gradient Boosting

Comparative Visualization:

- Comparisons of real AQI, predictions by the random forests, and Gradient Boosting predictions show:
- The models are useful in monitoring daily AQI shifts.
- Gradient Boosting is somewhat better than Random Forest in terms of its ability to follow steep AQI peaks.
- The scatter plots between each separate pollutant (PM2.5, NO 2, SO 2) and the predicted AQI confirm that the models are properly reflecting the impact of pollutants on the quality of the air.

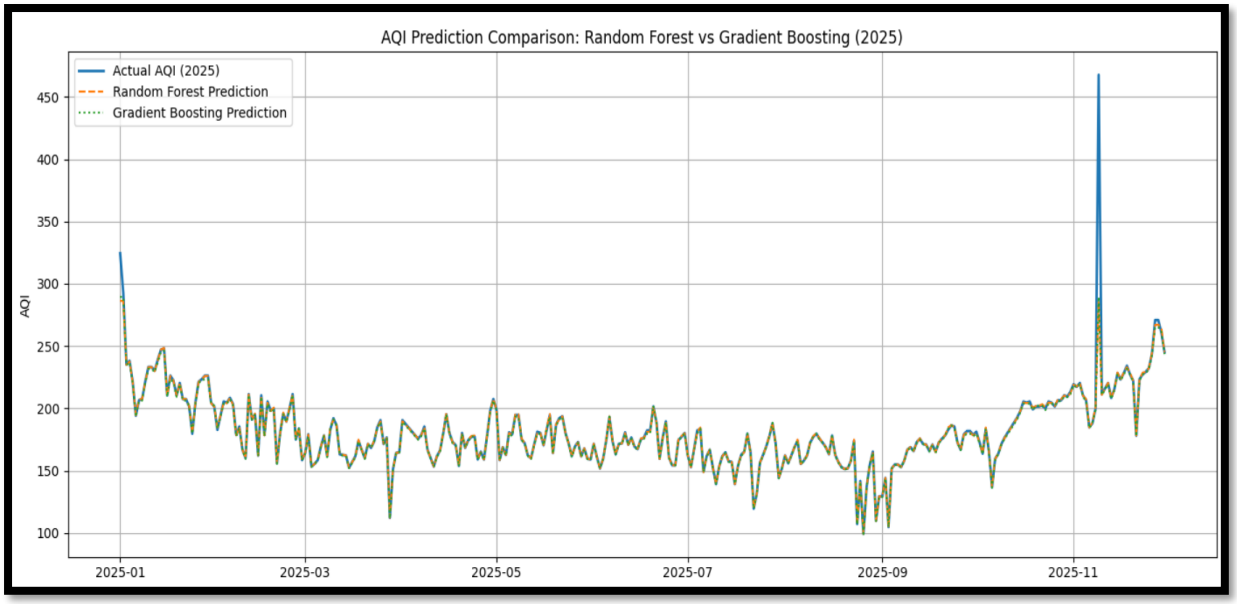


Figure 6: AQI Prediction Comparison: Random Vs Gradient Boosting

Random Forest: Actual vs Predicted

PM2.5 vs AQI

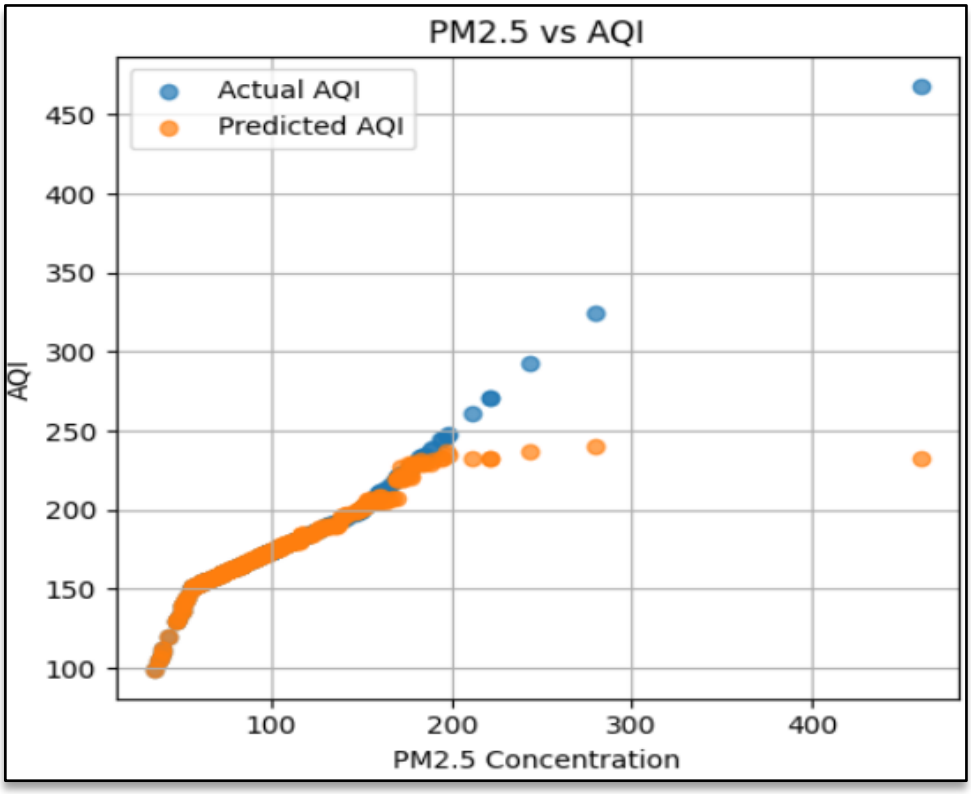


Figure 7 PM2.5 vs AQI

NO_2 vs AQI

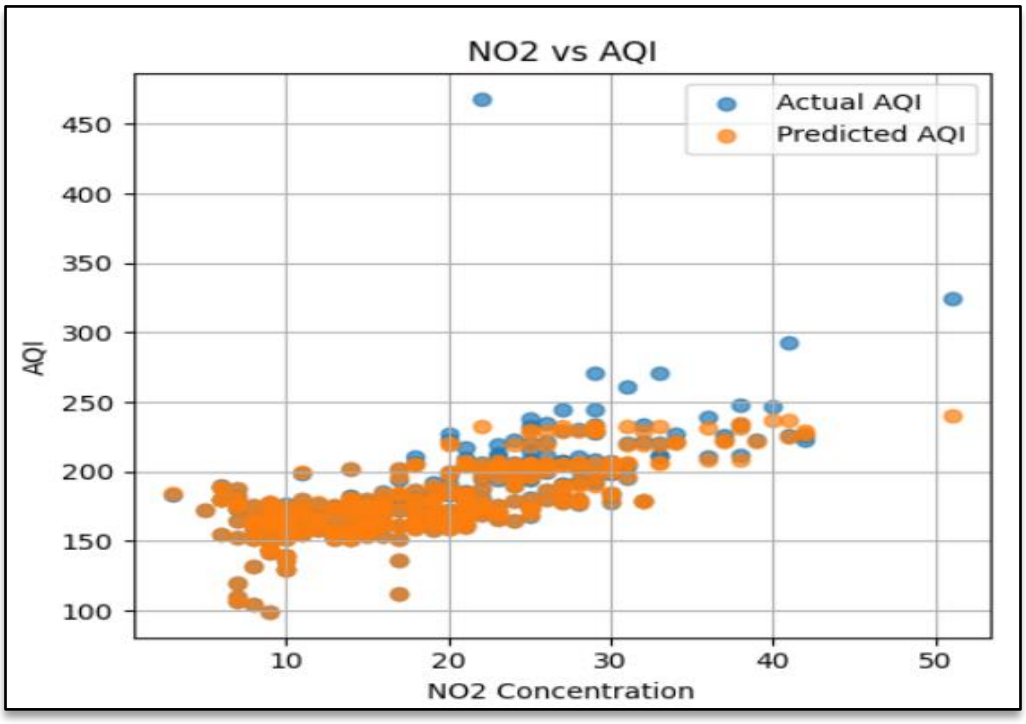


Figure 8 NO_2 vs AQI

SO_2 vs AQI

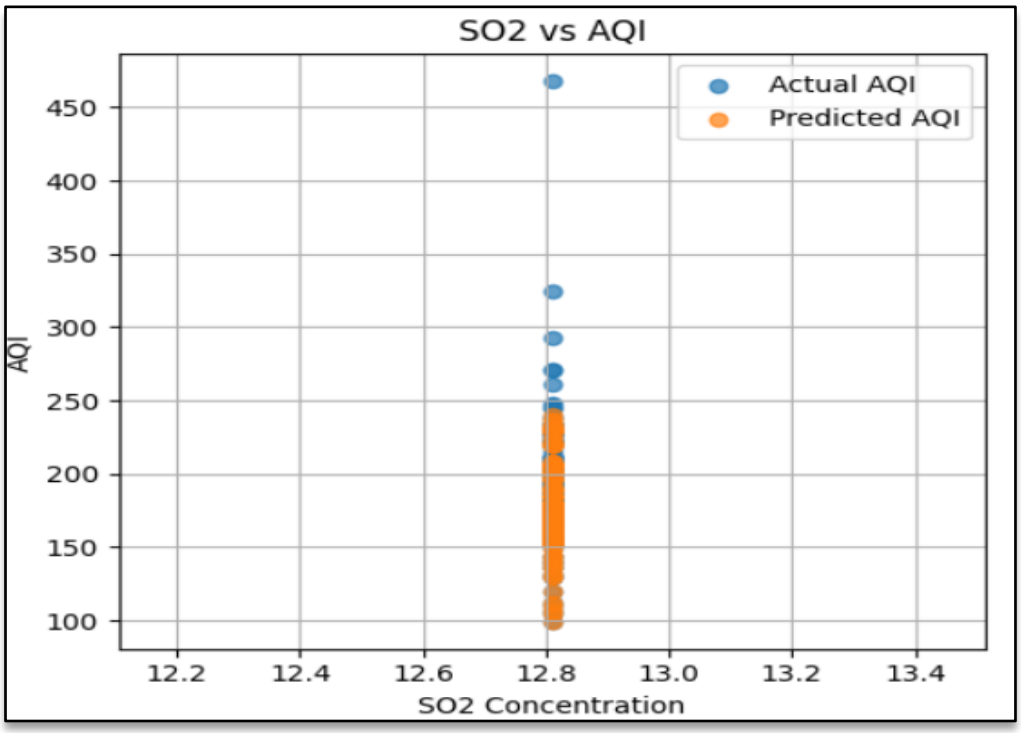


Figure 9 SO_2 vs AQI

Discussion:

The AQI prediction models were very effective in predicting the air quality. Both the Random Forest and Gradient Boosting were able to capture the day changes in pollutants though the latter was a little more accurate in the sharp peaks. The addition of lag-1 characteristics enabled the models to consider the effect of the past day pollution on the present AQI that enhanced the quality of the predictions. On the whole, the findings validate the hypothesis that past data on pollutants can be successfully applied in future air quality trend prediction.

Impact:

Proper prediction of AQI has important practical implications. The citizens are able to take early precautions when the pollution levels are high on a particular day thereby minimizing health risks. The government is able to anticipate interventions, warn and control emissions better. This predictive method assists in environmental administration, environmental health consciousness and decision guided by information, enhancing cleaner and safer urban areas.

Closing:

This project reveals a systematic method of predicting AQI with the help of Machine Learning. It offers robust and practical forecasts with the help of historical data, thorough preprocessing, feature engineering, and sophisticated modeling. The value of data-driven environmental monitoring envisioned in practice can be demonstrated through these predictions, which can, in turn, inform individuals and policymakers to enhance air quality and ensure the safety of the population.